David Jeehwan Lee

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Dual Citizenship (U.S. & Republic of Korea)

Education

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, South Korea
B.S. in Electrical Engineering, Cum Laude (GPA: 3.82/4.30), Completed in 3.5 years — *Mar. 2022 – Feb. 2027*

Daegu IL Science High School — Daegu, South Korea
High School Diploma, Valedictorian — *Mar. 2019 – Feb. 2022*

Research Experience

IMPACT Lab, KAIST — Jan. 2026 – Present
Undergraduate Researcher (Advisor: Prof. Minkyu Je) — *Daejeon, South Korea*

Quad-channel time-domain capacitance-to-digital converter (B.S. thesis)

- Showed with ISF-based time-variant simulations that asymmetric rise/fall times upconvert flicker noise and raise the $1/f^3$ corner, and that waveform symmetry suppresses it; moved to a relaxation oscillator for low jitter.
- Designed a quad-channel CDC for in-line microfluidic droplet sensing that places one sensor in each cell of a quadrature-phase swing-boosted RC oscillator, converting all four channels every period without multiplexing.
- Added flicker-aware gated-ring-oscillator TDCs with differenced dithering; in 0.18-μm, 0.9-V CMOS, simulations show 0.102 fF resolution in 5.9 μs at 110 μW and 0.040 fF calibrated four-channel error.

Flexible ion-gel pressure-sensor array readout (collaboration with IOEL, KAIST)

- Built a multiplexed readout for a 10×10 ion-gel capacitive sensor array: two MUXes route each row–column crossing to a PCap04 CDC, controlled by an Arduino over I^2C with a Python host for calibration and live display.
- Traced low readings to the sensor's high resistance and low frequency with LCR sweeps; slowing the CDC discharge doubled the pressure response, and calibrating out line parasitics enabled a real-time map at 7.7 frames/s.

Tearney Lab, Massachusetts General Hospital, Harvard-MIT HST — Jun. 2026 – Aug. 2026
Research Intern (Advisor: Prof. Guillermo J. Tearney) — *Boston, USA*

- Developed a 360° scanning μOCT probe for real-time deep brain stimulation (DBS) guidance that eliminates rotational distortion and fiber-optic rotary-joint loss by pairing a stationary extended-depth-of-field (EDOF) optical core with a rotating elliptical micro-mirror.
- Designed the optics in Zemax OpticStudio, fabricated the fiber core (cleaving, splicing, and tapering single- and multi-mode fiber), and integrated a Nanoscribe 3D-printed mirror into a handheld motorized prototype.
- Validated a 6–8 μm focal spot over a ∼200 μm depth of field with twist-free light delivery during rotation, and presented the work as a poster at the HST-Wellman Summer Institute.

AI-PIM Semiconductor Design Research Center, KAIST — Oct. 2025 – Dec. 2025
Research Intern (Funded by the Ministry of Science and ICT) — *Daejeon, South Korea*

- Ported an ASIC-targeted HyperRAM controller IP to an Artix-7 FPGA (Synplify, Vivado) to relieve the memory-bandwidth bottleneck of an AI SoC, resolving AXI4-to-HyperBus clock-domain crossings and tri-state I/O constraints; validated transactions on hardware with ILA.
- Integrated peripheral subsystems into a RISC-V AI SoC (clock/reset manager, AXI-to-APB bridges) and verified SPI microSD bring-up, DMA memory transfers, and PLIC/CLINT interrupt routing with bare-metal C tests.

Publications & Presentations

[1] **D. J. Lee**, W. Youn, and M. Je, "A Quad-Channel Simultaneous-Sampling CDC Based on a Quadrature-Phase Swing-Boosted RC Oscillator," *IEIE Fall Conference 2026, Undergraduate Paper Competition*, Nov. 2026.

[2] **D. J. Lee**, A. Ortega, H. Schulz-Hildebrandt, B. Davis, and G. J. Tearney, "Circumferential Scanning μOCT Probe for Real-Time Guidance in Deep Brain Stimulation," Poster, *HST-Wellman Summer Institute for Biomedical Optics*, Boston, MA, Aug. 2026.

Technical Projects

LSTM Breath Detection for a Self-Powered CO_2 Sensor Node | *Deep Learning* Jun. – Aug. 2025

- Ported a lab NDIR capnography chain (lock-in, temperature-compensated Beer–Lambert, end-tidal detection) from MATLAB to Python and built a simulator that turns 279 recorded exhalations into the node's 2 Hz CO_2 stream.
- Trained a 5,025-parameter LSTM that detected 87% of held-out walk-by exhales at 0.8 false alarms/h vs. 78% at 2.2/h for the best rule-based detector, with the gain on weak 120–250 ppm breaths (92% vs. 68%).
- Showed the model held F1 at 0.87 in an unseen 2,000 ppm room and would cost only 0.2–0.5% of the node's awake power, so inference can run on the sensor instead of a host.

TinyML Wake-Word Detection on a Raspberry Pi | *On-Device AI* Sep. 2025 – Dec. 2025

- Built an end-to-end wake-word pipeline (voice activity detection, MFCC, CNN) at 92.7% validation accuracy and tuned real-time streaming detection to a 90% hit rate with few false triggers.
- Found that INT8 quantization cut the model size by ∼40% but stalled the speedup on the CPU (unpacking overhead), showing that software compression alone hits a wall without matching hardware.

Leadership & Service

Reporter, EE Newsletter, KAIST School of Electrical Engineering Mar. 2025 – Present
English Mentor, KAIST Science Outreach Program (KSOP) Sep. 2025 – Jun. 2026
KAIST Student Representative (fully funded), ASPIRE League, Tokyo, Japan Jun. 2025
Korean Military Service, Capital Defense Command, Republic of Korea Army Aug. 2023 – Feb. 2025
Korean Student Representative (fully funded), KOFST Young Generation Forum Jul. 2023
Selected Trainee, Samsung Electronics Shining Star Program, Hwaseong Feb. 2023
Student Volunteer, Scientists and Engineers Without Borders, Myanmar Jan. 2018

Honors & Awards

Undergraduate Paper Award, The Institute of Electronics and Information Engineers (IEIE) 2026
Elite Soldier Award, Republic of Korea Army 2024
KAIST Science Gifted Student Scholarship, KAIST 2021
Dong-gu Education Foundation Scholarship, Daegu Metropolitan City 2021

Skills

Hardware: Embedded Systems (Raspberry Pi, Arduino), Digital SoC Design (Xilinx Vivado, Verilog), Analog Circuit Design (Cadence Virtuoso), Optical Design (Zemax OpticStudio)
Software: ML (PyTorch, TensorFlow), C/C++, Python, MATLAB
Certificate: Computer Utilization, Korea Chamber of Commerce and Industry
Languages: English (Native; TOEFL 118/120, TOEIC 990/990, OPIc AL), Korean (Native), French (A2)